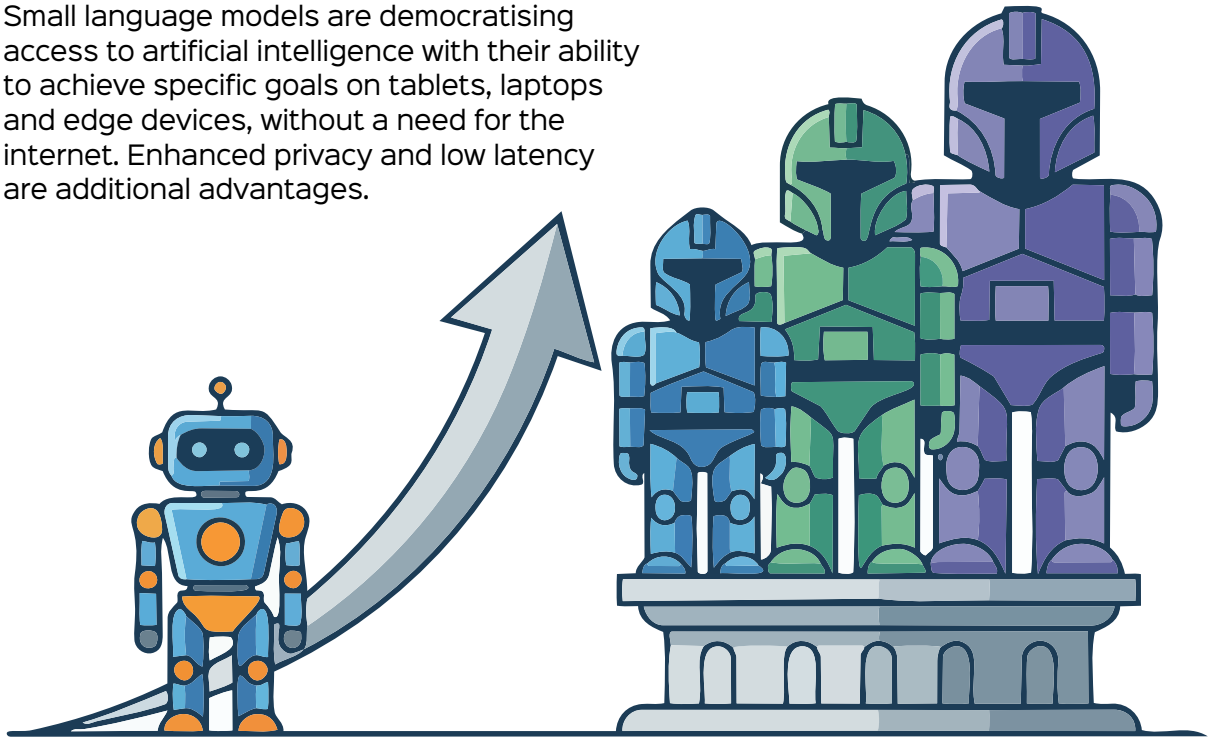


Small Language Models: The Rise of the Compact Titans

Small language models are democratising access to artificial intelligence with their ability to achieve specific goals on tablets, laptops and edge devices, without a need for the internet. Enhanced privacy and low latency are additional advantages.



Researchers at a university in a developing country, studying regional literature from past centuries, can now get the same results on an open model as they would have got from million-dollar data-centre access earlier.

This is not a forecast for 2035. This is true in February 2026!

The field of giant language models is currently experiencing a shift towards small language models (SLMs) that achieve near-state-of-the-art results while operating on mobile devices, computers, and edge devices. The products of these technologies operate at lower costs, deliver faster performance, consume less power, and provide users with greater control of their data.

Artificial intelligence has entered its phase of accessibility for everyone.

The dawn of compact AI

A doctor can today work at a rural clinic and use her smartphone to access a lightweight AI tool that gives her instant diagnostic results. Small language models (SLMs) are ‘compact titans’. They demonstrate that larger systems do not always deliver superior performance because their technology enables both multinational companies and independent researchers to access artificial intelligence.

SLMs have emerged as efficient underdogs. The Phi-3 model from Microsoft, the Gemma 2B model from Google, and Mistral’s products demonstrate that the path to success does not always need raw power — efficient design can get the same results. Model fine-tuning enables the achievement of advanced research

goals, while making artificial intelligence accessible even to users who have no technical expertise.

Decoding small language models

Think of large language models (LLMs) as massive cargo ships that have the power to transport extensive knowledge across the oceans yet operate with slow speed and high operational costs, and require excessive fuel consumption. SLMs resemble agile speedboats that possess the ability to navigate specific routes with exceptional speed because they operate with high agility and low fuel consumption. SLMs like Phi-3.5 (3.8B), Gemma 2 (2B-9B), Mistral Nemo (12B) and Qwen3 variants reach high performance levels and require fewer resources. The systems operate on laptops,